

Q18

Kadane's Algo:- Maxi subarray

nums = [-2, 1, -3, 4, -1, 2, 1, -5, 4]

Subarray

{-2}	{1}
{-2, 1}	{1, -3}
{-2, 1, -3}	{1, -3, 4}
{-2, 1, -3, 4}	{1, -3, 4, -1}
{-2, 1, -3, 4, -1}	{1, -3, 4, -1, 2}
{-2, 1, -3, 4, -1, 2}	{1, -3, 4, -1, 2, 1}
{-2, 1, -3, 4, -1, 2, 1}	{1, -3, 4, -1, 2, 1, -5}
{-2, 1, -3, 4, -1, 2, 1, -5}	{1, -3, 4, -1, 2, 1, -5, 4}
{-2, 1, -3, 4, -1, 2, 1, -5, 4}	↳ maxsum = 3
↳ maxsum = 4	

{-3}	{4}	{1}
{-3, 4}	{4, -1}	{1, 2}
{-3, 4, -1}	{4, -1, 2}	{1, 2, 1}
{-3, 4, -1, 2}	{4, -1, 2, 1}	{1, 2, 1, -5}
{-3, 4, -1, 2, 1}	{4, -1, 2, 1, -5}	{1, 2, 1, -5, 4}
{-3, 4, -1, 2, 1, -5}	{4, -1, 2, 1, -5, 4}	↳ maxsum = 1
{-3, 4, -1, 2, 1, -5, 4}	↳ maxsum = 5	
↳ maxsum = 2		

{2}	{4}	{-5}	{4}
{2, 1}	{1, -5}	{-5, 4}	↳ 4
{2, 1, -5}	{1, -5, 4}	↳ 1	
{2, 1, -5, 4}	↳ 0		
↳ max = 2			

nums = [-2, 1, -8, 4, -1, 2, 1, -5, 4]
 1 2 3 4 5 6 7 8

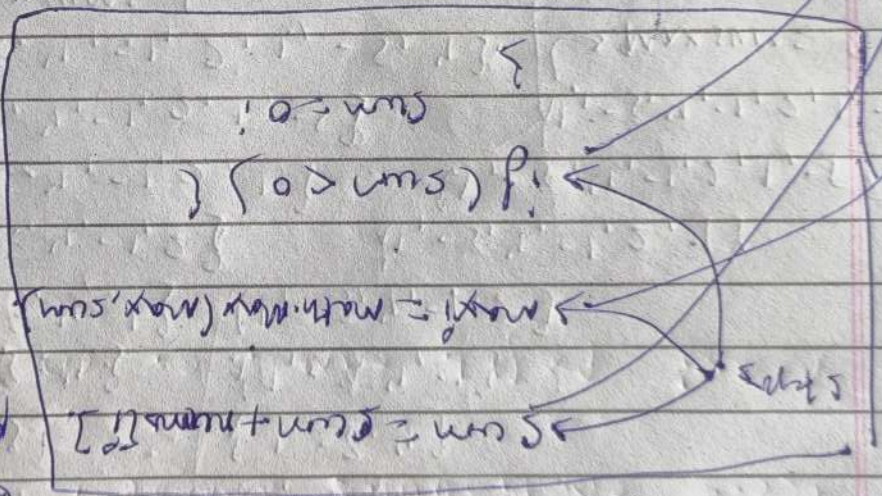
Initially

sum = 0
 max = -2 (let element be

equal right

problem me hoga
 hai, max

thodi nhi hoga



Main concept

He baadage ki current sum bithe hai

plus max ki update karega
 me se jo maximum hoga

Agar sum negative hai to sum
 ko 0 kar denge

Final Step

sum = 0, max = -2

i = 0, sum = 0 + (-2) = -2, max = max(-2, -2) = -2, sum = 0

i = 1, sum = 0 + (1) = 1, max = max(-2, 1) = 1, sum = 1

$i=2, \text{sum} = 1 + (-3) = -2, \text{maxi} = \max(1, -2) = 1, \text{sum} = 0,$

$i=3, \text{sum} = 0 + (4) = 4, \text{maxi} = \max(1, 4) = 4, \text{sum} = 4,$

$i=4, \text{sum} = 4 + (-1) = 3, \text{maxi} = \max(4, 3) = 4, \text{sum} = 3,$

$i=5, \text{sum} = 3 + (2) = 5, \text{maxi} = \max(4, 5) = 5, \text{sum} = 5,$

$i=6, \text{sum} = 5 + 1 = 6, \text{maxi} = \max(5, 6) = 6, \text{sum} = 6,$

$i=7, \text{sum} = 6 + (-5) = 1, \text{maxi} = \max(6, 1) = 6, \text{sum} = 1,$

$i=8, \text{sum} = 1 + 4 = 5, \text{maxi} = \max(6, 5) = 6, \text{sum} = 5,$

Final ans = 6

// code:-

```
int sum = 0;
```

```
int maxi = nums[0];
```

```
for(int i=0; i < nums.length; i++) {
```

```
    sum += nums[i];
```

```
    maxi = Math.Max(maxi, sum);
```

```
    if (sum < 0) {
```

```
        sum = 0;
```

```
    }
```

```
}
```

```
return maxi;
```

```
}
```

```
}
```

Time Complexity = $O(N)$.

Space Complexity = $O(1)$.

Think of Kadane's Algo when you see !-.

- Maximum sum Subarray
- Largest sum continuous segment
- Best continuous range.
- Maxi profit over a continuous period.